

Hurlstone Agricultural High School

Year 11

Assessment Task 1

Student Name: _____

Teacher Name: _____

2024

Mathematics Advanced

General Instructions

- **Senior Examiner:** Mrs M. Sabah
- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- NESA approved calculators may be used
- A Mathematics reference sheet is provided at the back of this paper
- For all Questions in Section II, show relevant mathematical reasoning and/or calculations

Total Marks:
45

Section I – 6 marks

- Attempt Questions 1 – 6
- Allow about 10 minutes for this section

Section II – 39 marks

- Attempt Questions 7 – 9 in the booklets provided
- Allow about 50 minutes for this section

Section I

6 marks

Attempt Questions 1 – 6

Allow about 10 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 6.

1. Which of the following is the solution to the equation $x^{-3} = \frac{x}{16}$ where $x \neq 0$?

A. $x = \pm 4$

B. $x = \pm 2$

C. $x = \pm \frac{1}{2}$

D. No Solution

2. Which of the following is the value of the expression $\frac{14^{n+4} + 7^{n+3} \times 2^{n+3}}{13 \times 14^n + 14^n}$?

A. 2 940

B. 2 049

C. 2 490

D. 2 094

3. Which of the following equations is equivalent to $y = \frac{3}{4}x + \frac{1}{2}$?

A. $3x - 4y - 2 = 0$

B. $3x + 4y + 2 = 0$

C. $6x - 8y - 2 = 0$

D. $3x - 4y + 2 = 0$

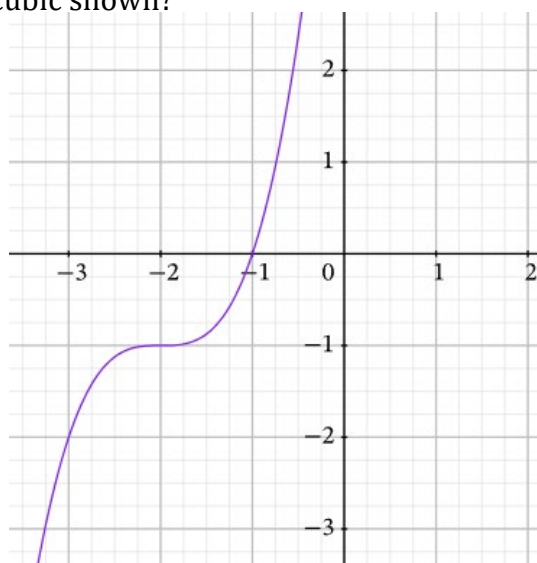
4. Which of the following represents the equation of the cubic shown?

A. $y = (x - 2)^3 - 1$

B. $y = (x + 2)^3 - 1$

C. $y = (x + 2)^3 + 1$

D. $y = (x - 2)^3 + 1$



Section I continues on next page ...

Section I continued ...

5.

Consider $f(x) = \frac{1}{1+x}$, then which of the following is equal to $f(f(x))$?

A. $\frac{1}{(1+x)^2}$

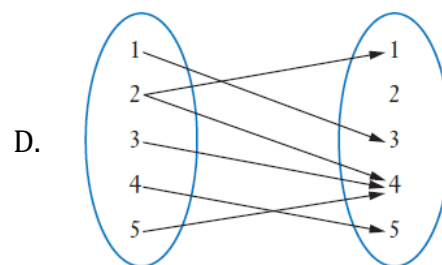
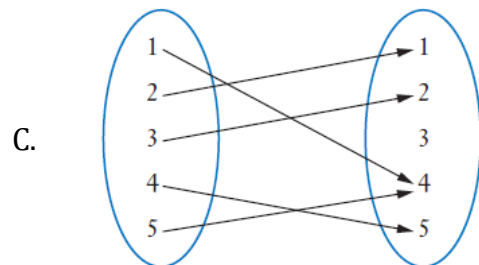
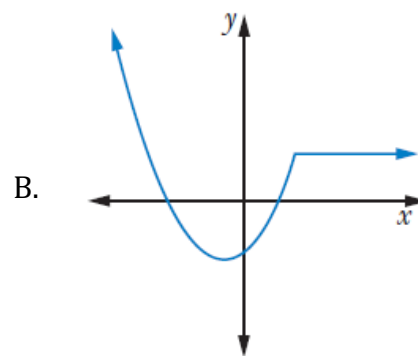
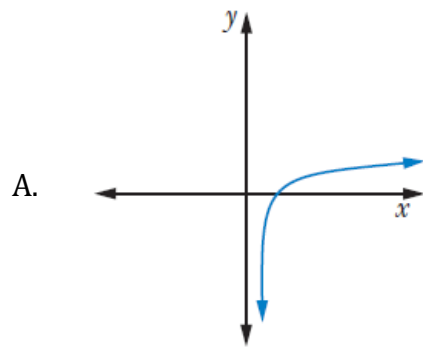
B. $\frac{1+x}{2+x}$

C. 1

D. $\frac{1}{2+x}$

6.

Which of the following represents a one-to-one function?



~ End of Section I ~

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

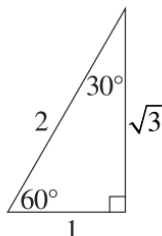
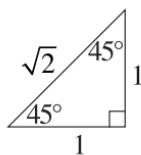
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

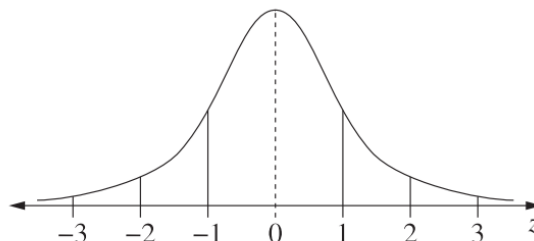
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= {}^n C_x p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}_nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

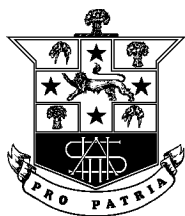
$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

BLANK PAGE



Students Name: _____

Teachers Name: _____

2024

Hurlstone Agricultural High School
Year 11
Assessment Task 1

Mathematics Advanced
Multiple-Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A ☒ B ☐ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

Question

1.	A ☐	B ☐	C ☐	D ☐
2.	A ☐	B ☐	C ☐	D ☐
3.	A ☐	B ☐	C ☐	D ☐
4.	A ☐	B ☐	C ☐	D ☐
5.	A ☐	B ☐	C ☐	D ☐
6.	A ☐	B ☐	C ☐	D ☐



Students Name: _____

Teachers Name: _____

2024

Hurlstone Agricultural High School

Year 11

HSC Assessment Task 1

13

Mathematics Advanced

Section II

Question 7

(13 Marks)

Algebraic Techniques

- Answer each question in the space provided.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 7

- (a) Solve the quadratic equation $10 + 6x - x^2 = 0$ by completing the square.

2

- (b) Show that the expression below simplifies to a positive integer, stating its value.

$$\frac{1 + \sqrt{7}}{3 - \sqrt{7}} - \frac{8 - \sqrt{7}}{\sqrt{7} - 2}$$

3

Question 7 continued ...

- (c) Solve the equation $3x(x + 4) = -3x + 4$, giving your answer to 2 decimal places. **2**

- (d) Factorise the expression $2b^3 + 3a^3 + 3ab^2 + 2a^2b$. 2

Question 7 continued ...

(e) Given

$$y = \frac{2m + 4}{m^2 - 25} \times \frac{m + 5}{m^2 + 3m + 2} \div \frac{2}{3m^2 - 16m + 5}$$

(i) Show that the equation above simplifies to $y = \frac{3m - 1}{m + 1}$. 2

.....

.....

.....

.....

.....

.....

.....

.....

(ii) Make m the subject of the equation from (i). 2

.....

.....

.....

.....

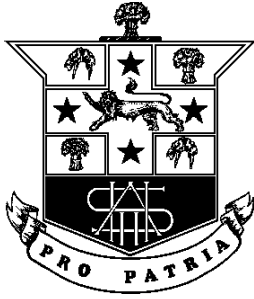
.....

.....

.....

.....

~ End of Question 7 ~



Students Name: _____

Teachers Name: _____

2024

Hurlstone Agricultural High School

Year 11

HSC Assessment Task 1

13

Mathematics Advanced

Section II

Question 8

(13 Marks)

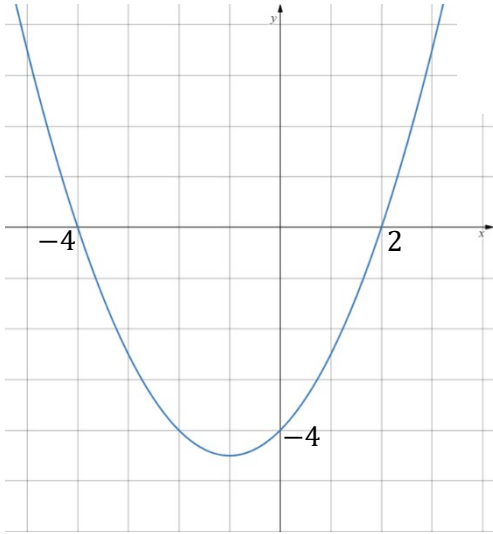
Linear, Quadratic and Cubic Functions

- Answer each question in the space provided.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 8

(a) Find the equation of the parabola shown in the graph below.

2



.....

.....

.....

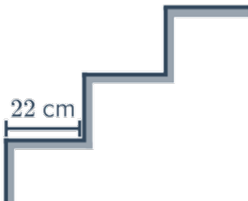
.....

.....

.....

(b) A Glenfield station staircase is built so that its maximum steepness (gradient) is 0.6. Each step goes in by 22 cm. Yashika counted 31 steps from the platform to the concourse.

What is the vertical height, in metres, of the staircase?



2

.....

.....

.....

.....

.....

Question 8 continued ...

- (c) The height in metres, h , of a golf ball is given by the equation $h = -16.1t^2 + 137t + 3$, where t is the time in seconds after it was struck.
Jammy hit a golf ball modelling the equation.
He wins a prize if the ball reaches a height of 301 m.
Does Jammy win a prize? Justify your answer, include calculations.

2

.....

.....

.....

.....

.....

.....

- (d) Solve the equations $y = x + 1$ and $x^2 + y^2 = 13$ simultaneously.

2

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Question 8 continued ...

- (e)

A quadratic function has the equation $y = x^2 + 6x - 7$.
Find the coordinates of the vertex.

1

.....

.....

.....

.....

.....

- (f)

The point A has coordinates $(-2, 7)$ and the point B has coordinates $(4, p)$.
The point M is the midpoint of AB and has coordinates $(q, \frac{9}{2})$.

- (i)

find the values of p and q .

2

.....

.....

.....

.....

- (ii)

find the equation, in general form, of the perpendicular bisector of AB .

2

.....

.....

.....

.....

.....

.....

~ End of Question 8 ~



Students Name: _____

Teachers Name: _____

2024

Hurlstone Agricultural High School

Year 11

HSC Assessment Task 1

13

Mathematics Advanced

Section II

Question 9

(13 Marks)

Introduction to Functions

Further Functions and Relations

- Answer each question in the space provided.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 9

(a) Given that $f(x) = 2x + 3$ and $g(x) = 5x + b$.

Find the value of b given that $g(f(x)) = f(g(x))$. **2**

.....

.....

.....

.....

.....

.....

(b) Solve $|-6x + 3| = 27$. **2**

.....

.....

.....

.....

.....

.....

(c) Determine if the function $f(x) = -3x^3 + 2x$ is even, odd or neither. **2**

.....

.....

.....

.....

.....

.....

Question 9 continued ...

(d)

Consider the function $f(x) = |x - 3| - 2$.

(i) Sketch the graph of $f(x)$, showing all important features.

2

(ii) State the domain and range of $f(x)$ in interval notation.

2

.....

.....

.....

(e)

State the domain and range of $h(x) = \sqrt{9 - x^2}$.

1

.....

.....

.....

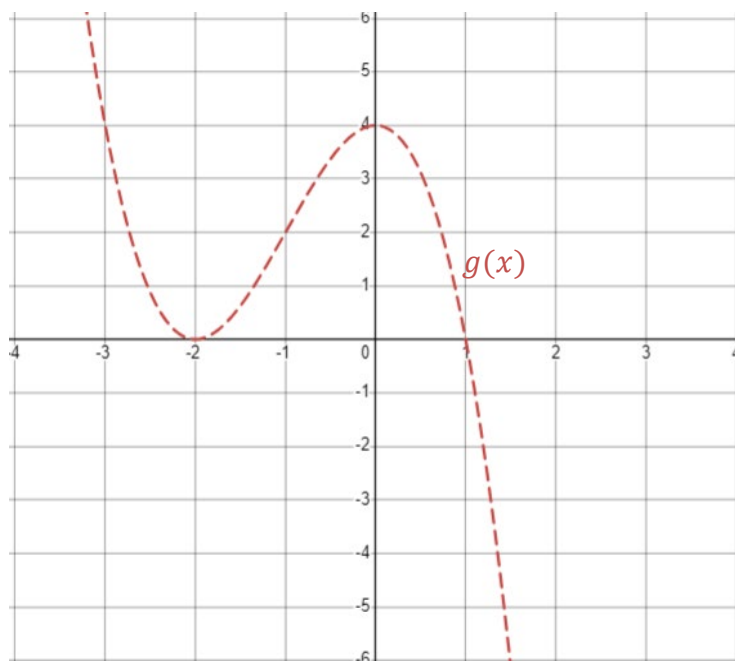
Question 9 continued ...

(f)

The graph below shows the curve $y = g(x)$.

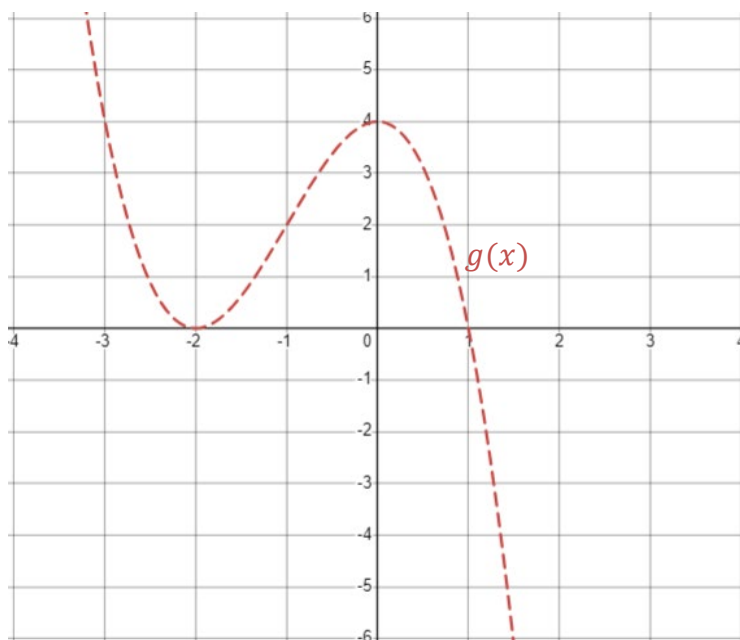
(i) Sketch the graph of $y = g(x - 2)$, showing all important features.

1



(ii) Sketch the graph of $y = -g(x)$, showing all important features.

1



~ End of Question 9 ~

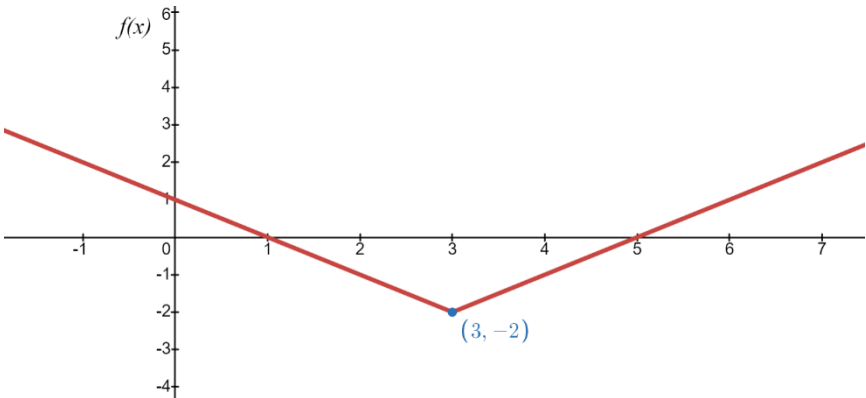
END OF TASK 1

Year 11		Mathematics Advanced	2024 Task 1
Multiple Choice Qs		Solutions and Marking Guidelines	
Outcomes Addressed in this Question			
MA11-2 uses the concepts of functions and relations to model, analyse and solve practical problems			
Question	Solutions	Marking Guidelines	
1	$x^{-3} = \frac{x}{16}$ $\frac{1}{x^3} = \frac{x}{16}$ $x^4 = 16$ $x = \pm 2$ Answer: B		
2	$\frac{14^{n+4} + 7^{n+3} \times 2^{n+3}}{13 \times 14^n + 14^n}$ $= \frac{14^{n+4} + (7 \times 2)^{n+3}}{14^n(13 + 1)}$ $= \frac{14^{n+4} + 14^{n+3}}{14^n(13 + 1)}$ $= \frac{14^n \times (14^4 + 14^3)}{14^n(14)}$ $= \frac{(14^4 + 14^3)}{(14)} = 2940$ Answer: A		
3	$y = \frac{3}{4}x + \frac{1}{2}?$ $8y = 6x + 4$ $4y = 3x + 2$ $0 = 3x - 4y + 2$ $\therefore D$		
4	The cubic $y = x^3$ has been translated 2 places to the left and 1 place down. $\therefore y = (x + 2)^3 - 1$ $\therefore B$		
5	B		
6	A		

Year 11	Mathematics Advanced 2024	TASK 1
Solutions and Marking Guidelines		
Outcomes Addressed in this Question		
MA11-1 uses algebraic and graphical techniques to solve, and where appropriate, compare alternative solutions to problems		
Part / Outcome	Solutions	Marking Guidelines
ER Q7(a) MA11-1	$10 + 6x - x^2 = 0$ $10 + 6x - x^2 = 0$ $x^2 - 6x - 10 = 0$ $x^2 - 6x + (-3)^2 - 9 - 10 = 0$ $(x - 3)^2 = 19$ $x - 3 = \pm\sqrt{19}$ $x = 3 \pm \sqrt{19}$	2 marks – Correct solution 1 mark – Substantially correct
ER Q7 (b) MA11-1	$\frac{1 + \sqrt{7}}{3 - \sqrt{7}} - \frac{8 - \sqrt{7}}{\sqrt{7} - 2}$ $= \frac{1 + \sqrt{7}}{3 - \sqrt{7}} \times \frac{3 + \sqrt{7}}{3 + \sqrt{7}} - \frac{8 - \sqrt{7}}{\sqrt{7} - 2} \times \frac{\sqrt{7} + 2}{\sqrt{7} + 2}$ $= \frac{3 + \sqrt{7} + 3\sqrt{7} + 7}{9 - 7} - \frac{8\sqrt{7} + 16 - 7 - 2\sqrt{7}}{7 - 4}$ $= \frac{10 + 4\sqrt{7}}{2} - \frac{6\sqrt{7} + 9}{3}$ $= 5 + 2\sqrt{7} - 2\sqrt{7} - 3 = 2$	3 marks – Correct solution 2 marks – Substantially correct solution. 1 mark – some correct working towards correct solution
ER Q7 (c) MA11-1	$3x(x + 4) = -3x + 4$ $3x^2 + 15x - 4 = 0$ $x = \frac{-15 \pm \sqrt{15^2 - 4(3)(-4)}}{2(3)}$ $x = 0.25 \text{ or } -5.25 \text{ (to 2 decimal places)}$	2 marks – Correct solution 1 mark – Substantially correct
ER Q7 (d) MA11-1	$2b^3 + 3ab^2 + 3a^3 + 2a^2b$ $= b^2(2b + 3a) + a^2(3a + 2b)$ $= (2b + 3a)(a^2 + b^2)$	2 marks – Correct solution 1 mark – Substantially correct
ER Q7 (e)(i) MA11-1	$y = \frac{2m + 4}{m^2 - 25} \times \frac{m + 5}{m^2 + 3m + 2} \div \frac{2}{3m^2 - 16m + 5}$ $y = \frac{2(m + 2)}{(m - 5)(m + 5)} \times \frac{m + 5}{(m + 1)(m + 2)} \times \frac{(3m - 1)(m - 5)}{2}$	2 marks – Correct solution

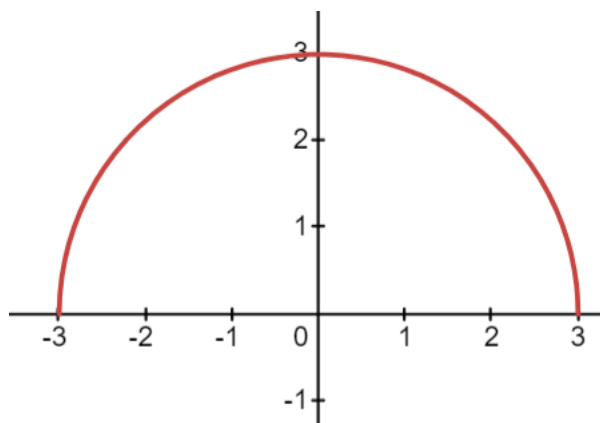
Year 11		Mathematics Advanced	2024 Task 1
Solutions and Marking Guidelines			
Outcomes Addressed in this Question			
MA11-2 uses the concepts of functions and relations to model, analyse and solve practical problems			
Question	Solutions		Marking Guidelines
Question 8			
a.	$y = a(x + 4)(x - 2)$ Point $(0, -4)$ lies on the parabola $(-4) = a((0) + 4)((0) - 2)$ $-4 = -8a$ $\frac{1}{2} = a$ $\therefore$ equation of parabola is $y = \frac{1}{2}(x + 4)(x - 2)$	2 marks: correct solution 1 mark: significant progress towards a correct solution	
b.	Gradient of each step = $\frac{\text{rise}}{\text{run}}$ Let height of each step = h $0.6 = \frac{h}{22}$ $h = 13.2 \text{ cm}$ $\therefore$ height of vertical staircase is = 13.2×31 = 409.2 m	2 marks: correct solution 1 mark: significant progress towards a correct solution	
c	$h = -16.1t^2 + 137t + 3$ $301 = -16.1t^2 + 137t + 3$ $16.1t^2 - 137t + 298 = 0$ $\Delta = b^2 - 4ac$ $\Delta = (-137)^2 - 4(16.1)(298)$ $\Delta = -422.2 < 0$ So equation has no solution $\therefore$ Jammy did not win a prize	2 marks: correct solution 1 mark: significant progress towards a correct solution	
d	$y = x + 1 \dots\dots\dots (1)$ $x^2 + y^2 = 13 \dots\dots\dots (2)$ Sub (1) into (2): $x^2 + (x + 1)^2 = 13$ $x^2 + x^2 + 2x + 1 = 13$ $2x^2 + 2x - 12 = 0$ $x^2 + x - 6 = 0$ $(x + 3)(x - 2) = 0$ $x = 2 \text{ or } x = -3$ When $x = 2$, $y = 3$ When $x = -3$, $y = -2$ $\therefore$ Solutions are $x = 2$, $y = 3$ and $x = -3$, $y = -2$	2 marks: correct solution 1 mark: significant progress towards a correct solution	

Question		
e	$y = x^2 + 6x - 7$ Axis of symmetry: $x = -\frac{b}{2a}$ $x = -\frac{6}{2(1)} = -3$ When $x = -3, y = -16$. $\therefore$ Vertex $(-3, -16)$	1 mark: correct solution
f i)	$M\left(q, \frac{9}{2}\right) = \left(\frac{-2+4}{2}, \frac{7+p}{2}\right)$ $M\left(q, \frac{9}{2}\right) = \left(1, \frac{7+p}{2}\right)$ $q = 1, \quad \frac{9}{2} = \frac{7+p}{2}$ $18 = 14 + 2p$ $4 = 2p$ $p = 2$	2 marks: correct solution 1 mark: significant progress towards a correct solution
f ii)	$M\left(1, \frac{9}{2}\right), \quad A(-2, 7), \quad B(4, 2)$ $m_{AB} = \frac{2-7}{4-(-2)}$ $m_{AB} = -\frac{5}{6}$ Gradient of perpendicular bisector $= \frac{6}{5}$ It passes through the midpoint $M\left(1, \frac{9}{2}\right)$ $y - y_1 = m(x - x_1)$ $y - \frac{9}{2} = \frac{6}{5}(x - 1)$ $10y - 45 = 12(x - 1)$ $10y - 45 = 12x - 12$ $0 = 12x - 12 - 10y + 45$ $\therefore 12x - 10y + 33 = 0$	2 marks: correct solution 1 mark: significant progress towards a correct solution

Year 11	Mathematics Advanced	Assessment Task 1 2024
Question No. 9	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
MA11-2 uses the concepts of functions and relations to model, analyse and solve practical problems		
Part	Solutions	Marking Guidelines
(a)	$g(f(x)) = f(g(x))$ $g(2x + 3) = f(5x + b)$ $5(2x + 3) + b = 2(5x + b) + 3$ $10x + 15 + b = 10x + 2b + 3$ $12 = b$ $b = 12$	2 marks For correct solution 1 mark Substantial progress towards the correct solution.
(b)	$ -6x + 3 = 27$ $-(-6x + 3) = 27 \quad \text{or} \quad +(-6x + 3) = 27$ $6x - 3 = 27 \quad \text{or} \quad -6x + 3 = 27$ $6x = 30 \quad \text{or} \quad -6x = 24$ $x = 5 \quad \text{or} \quad x = -4$	2 marks For correct solution 1 mark Substantial progress towards the correct solution.
(c)	Consider $f(-x) = -3(-x)^3 + 2(-x)$ $= -3(-x^3) - 2x$ $= 3x^3 - 2x$ $= -(-3x^3 + 2)$ $\therefore f(-x) = -f(x)$ $\therefore f(x)$ is an ODD function	2 marks For correct solution 1 mark Substantial progress towards the correct solution.
(d)	$f(x) = x - 3 - 2$	2 marks For correct solution
(i)		1 mark Substantial progress towards the correct solution.
(ii)	Domain: $(-\infty, \infty)$ Range: $[-2, \infty)$	2 marks For correct solution 1 mark Substantial progress towards the correct solution.

(e)

$$h(x) = \sqrt{9 - x^2}$$



Domain: $-3 \leq x \leq 3$ or $|x| \leq 3$ or $[-3, 3]$

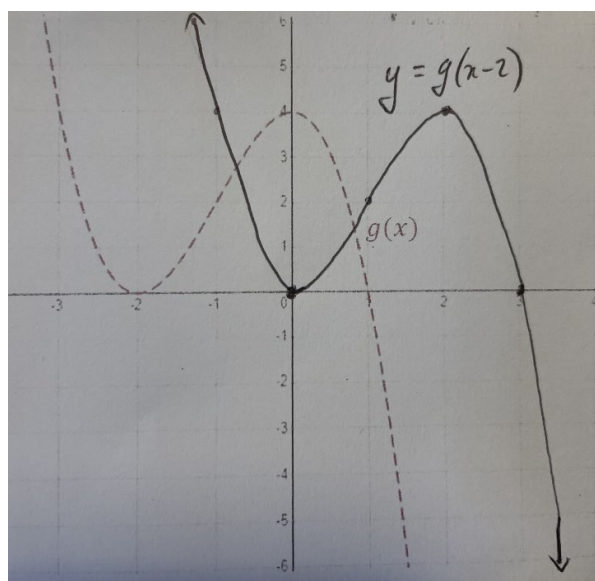
Range: $0 \leq h(x) \leq 3$ or $[0, 3]$

1 mark

For correct solution

(f)

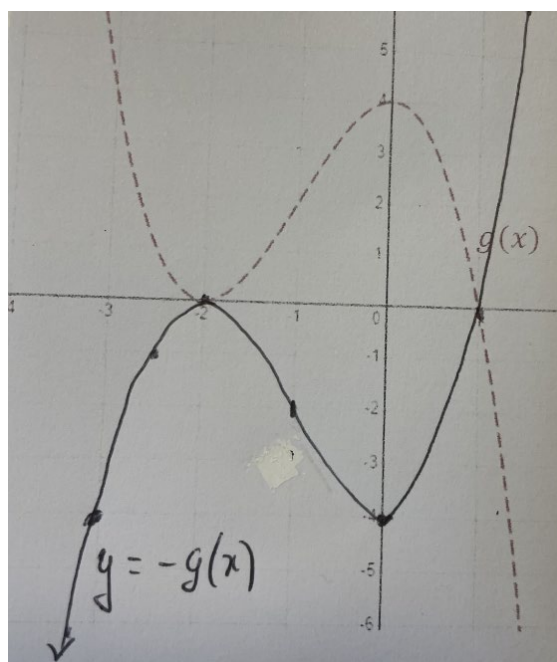
(i)



1 mark

For correct solution

(ii)



1 mark

For correct solution